

[T]-Theory: Physics

P2 · Formal Foundation · For: The Astrophysicist

The Master Green's Function — from Helmholtz to the Hubble horizon.

Cosmological predictions (zero-parameter):

- $\Omega_\Lambda = 7/11 \approx 0.636$ observed: 0.683 (6.8% off)
- $\Omega_c = 3/11 \approx 0.273$ observed: 0.265 (2.9% off)
- $\Omega_b = 1/22 \approx 0.045$ observed: 0.049 (8% off)
- $w = -1$ exact; live test: DESI DR1 & Euclid

Derived, not postulated:

- Λ from compact vacuum fraction $7/11$
- Dark matter = spatial block vacuum — no WIMPs, no direct detection
- Cosmological constant problem resolved topologically
- Moduli frozen by G_2 holonomy (LocalGR gate theorem, Lean 4)

I · The Master Green's Function

The USF propagator in flat 4D spacetime (Lorenz gauge, retarded):

$$G_R(r, \tau) = \frac{v_s}{4\pi r} e^{-kv_s\tau} \delta\left(\tau - \frac{r}{v_s}\right) \theta(\tau)$$

The **Euclidean QFT formulation** (Osterwalder-Schrader, P14–P15):

$$S[\Phi] = \int d^4x \left[\frac{1}{2}(\partial_\mu \Phi)^2 + \frac{1}{2}k^2\Phi^2 + \frac{\lambda}{4!}\Phi^4 \right]$$

All four OS axioms verified in Lean 4 (P14). Reflection positivity confirmed under Hopfield quartic coupling (P15).

II · The M-Theory Geometry

$$M_{11} = \mathbb{R}_t \times M_3 \times X_7, \quad X_7 = CY_3 \times S^1/\mathbb{Z}_2$$

The vacuum energy partition from dimension counting alone:

$$\Omega_\Lambda = \frac{\dim X_7}{\dim M_{11}} = \frac{7}{11}, \quad \Omega_c = \frac{\dim M_3}{\dim M_{11}} = \frac{3}{11}$$

G_2 holonomy on X_7 freezes the moduli: $d\Omega_\Lambda/dz = 0$.

Proved in Lean 4 via the LocalGR gate theorem chain.

III · Scale 19–20 on the Dial

At cosmological scale ($n = 19$), the field equation reduces to the linearised Einstein equation. At $n = 20$ (Hubble scale), the propagator becomes:

$$G_{20}(x, x') = \text{gravitational wave propagator} = \frac{e^{-m_g r}}{r}$$

with effective graviton mass $m_g \rightarrow 0$ on the Hubble horizon.

IV · Key Theorems (Lean 4, 0 sorries)

- `cosmological_correspondence`: scale 19 = linearised GR ✓
- `calabi_yau_moduli_static`: $d\Omega_\Lambda/dz = 0$ ✓
- `dm_gauge_coupling_zero`: DM has zero EM coupling ✓
- `usf_all_predictions_within_bounds`: all three Planck hits ✓

G-ID: *The Master Green's Function* — $(\nabla^2 + k^2)G = \delta$ in relativistic field theory. ORCID: 0009-0007-2194-0850 · CC BY 4.0 · Zurich 2026